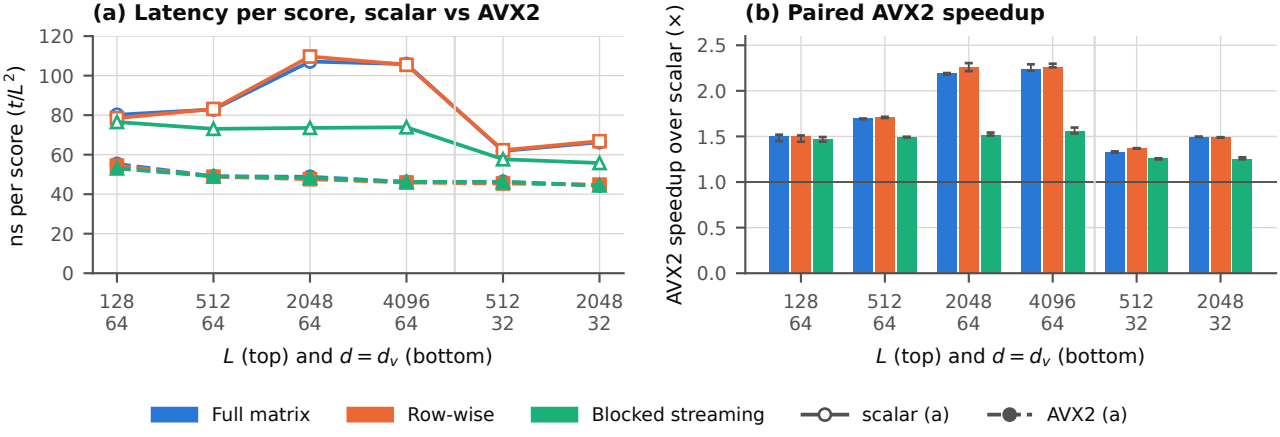
(a) Latency per score, scalar vs AVX2
ns per score (t/L^2)
120
100
80
60
40
20
0
128
64
512
64
2048
64
4096
64
512
32
2048
32
L (top) and $d = d_v$ (bottom)
(b) Paired AVX2 speedup
AVX2 speedup over scalar (×)
2.5
2.0
1.5
1.0
0.5
0.0
128
64
512
64
2048
64
4096
64
512
32
2048
32
L (top) and $d = d_v$ (bottom)
Full matrix
Row-wise
Blocked streaming
scalar (a)
AVX2 (a)